

Paideia Math Team Practice

Triangles: Heron's Formula and Shoelace / Area Methods

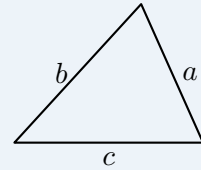
Heron's Formula

For a triangle with side lengths a , b , and c , let the semiperimeter be

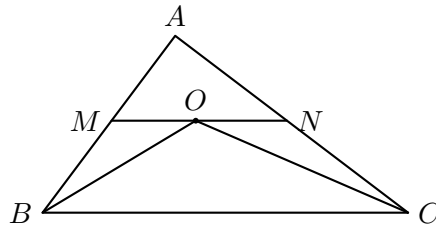
$$s = \frac{a + b + c}{2}.$$

Then the area is

$$A = \sqrt{s(s-a)(s-b)(s-c)}.$$



In the adjoining diagram, BO bisects $\angle CBA$, CO bisects $\angle ACB$, and MN is parallel to BC . If $AB = 12$, $BC = 24$, and $AC = 18$, then the perimeter of $\triangle AMN$ is



- A. 30 B. 33 C. 36 D. 39 E. 42

Triangle ABC has $AB = 27$, $AC = 26$, and $BC = 25$. Let I denote the intersection of the internal angle bisectors of $\triangle ABC$. What is BI ?

- A. 15 B. $5 + \sqrt{26} + 3\sqrt{3}$
 C. $3\sqrt{26}$ D. $\frac{2}{3}\sqrt{546}$
 E. $9\sqrt{3}$

A *disphenoid* is a tetrahedron whose triangular faces are congruent to one another. What is the least total surface area of a disphenoid whose faces are scalene triangles with integer side lengths?

- A. $\sqrt{3}$ B. $3\sqrt{15}$ C. 15 D. $15\sqrt{7}$ E. $24\sqrt{6}$

Shoelace Method

For a polygon whose vertices are listed in order around the boundary,

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n),$$

repeat the first vertex at the end. Multiply diagonally down to the right and add; then multiply diagonally down to the left and add. The area is one-half the absolute value of the difference:

$$A = \frac{1}{2} |(x_1y_2 + x_2y_3 + \dots + x_ny_1) - (y_1x_2 + y_2x_3 + \dots + y_nx_1)|.$$

A useful setup is

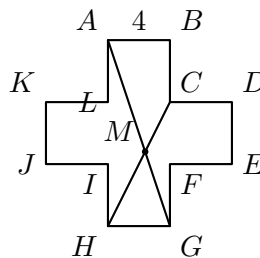
x	y	
x_1	y_1	
x_2	y_2	$\searrow$ add these products
$\vdots$	$\vdots$	
x_n	y_n	$\nearrow$ add these products
x_1	y_1	

The vertices must be taken consecutively clockwise or counterclockwise around the polygon.

Quadrilateral $ABCD$ is a parallelogram, and E is the midpoint of the side AD . Let F be the intersection of lines EB and AC . What is the ratio of the area of quadrilateral $CDEF$ to the area of triangle CFB ?

- A. 5 : 4 B. 4 : 3 C. 3 : 2 D. 5 : 3 E. 2 : 1

Consider the 12-sided polygon $ABCDEFGHIJKL$, as shown. Each of its sides has length 4, and each two consecutive sides form a right angle. Suppose that AG and CH meet at M . What is the area of quadrilateral $ABCM$?



- A. $\frac{44}{3}$ B. 16 C. $\frac{88}{5}$ D. 20 E. $\frac{62}{3}$

Suppose z is a complex number with positive imaginary part, with real part greater than 1, and with $|z| = 2$. In the complex plane, the four points $0, z, z^2$, and z^3 are the vertices of a quadrilateral with area 15. What is the imaginary part of z ?

- A. $\frac{3}{4}$ B. 1 C. $\frac{4}{3}$ D. $\frac{3}{2}$ E. $\frac{5}{3}$